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Started	Thursday, 15 October 2026, 8:42 PM
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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Combining sorted sublists
- c. Sorting the entire list sequentially
- d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. sort()
- c. order()
- d. arrange()

The correct answer is: sort()

Question **3**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Quick sort
- b. Heap sort
- c. Bubble sort
- d. Insertion sort

The correct answer is:
Insertion sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Finding a specific element in a list
- c. Merging two datasets
- d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **5**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It makes the analysis slower
- b. It has no effect on the analysis process
- c. It complicates the analysis process
- d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **6**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation
b. It increases the number of comparisons needed
c. It makes the search operation slower
d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Quick Sort
b. Bubble Sort
c. Heap Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Selection
- b. All of the above
- c. Searching
- d. Duplicates

The correct answer is: All of the above

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Merge Sort
c. Insertion Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time
b. It divides the list into two halves, sorts each half, and then merges them
c. It compares adjacent elements and swaps them if necessary
d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **12**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems
b. It divides the list into sublists
c. It uses the heap data structure
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **13**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. order()
b. arrange()
c. sorted()
d. sort()

The correct answer is: sorted()

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It only works on small datasets
- c. It avoids the need for recursion
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **15**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes the data unsorted
- b. It complicates the selection of items
- c. It makes it easier to select items based on their relationship to the rest of the items
- d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items